

AGE 200 INTRODUCTION TO QUANTITATIVE AND QUALITATIVE METHODS IN STATISTICS

Chapter # 03

Measures of Central Tendency

Measure of Central Tendency: Usually when two or more different data sets are to be compared it is necessary to condense the data, but for comparison the condensation of data set into a frequency distribution and visual presentation are not enough. It is then necessary to summarize the data set in a single value. Such a value usually somewhere in the center and represent the entire data set and hence it is called measure of central tendency or averages. Since a measure of central tendency (i.e. an average) indicates the location or the general position of the distribution on the X-axis therefore it is also known as a measure of location or position.

Types of Measure of Central Tendency

1. Arithmetic Mean
2. Geometric Mean
3. Harmonic Mean
4. Mode
5. Median

Arithmetic Mean or Simply Mean: *“A value obtained by dividing the sum of all the observations by the number of observation is called arithmetic Mean”*

$$\text{Mean} = \frac{\text{Sum of All observation}}{\text{Number of observation}}$$

Methods	Ungrouped data	Grouped data
Direct Method	$\bar{x} = \frac{\sum x_i}{n}$	$\bar{x} = \frac{\sum fx}{n}$; Here $n = \sum f$
Short cut Method	$\bar{x} = A + \frac{\sum D}{n}$	$\bar{x} = A + \frac{\sum fD}{n}$; Here $n = \sum f$
	Where $D = X_i - A$ and A is the provisional or assumed mean.	
Step deviation Method	$\bar{x} = A + \frac{\sum u}{n} \times h$	$\bar{x} = A + \frac{\sum fu}{n} \times h$; Here $n = \sum f$
	Where $u = \frac{X_i - A}{h}$ and h is the common width of the class intervals	

Numerical Example:

Calculate the arithmetic mean for the following the marks obtained by 9 students are given below:

Using formula of arithmetic mean for ungrouped data:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$n = 9$$

$$\bar{x} = \frac{360}{9} = 40 \text{ marks}$$

x_i
45
32
37
46
39
36
41
48
36
$\sum_{i=1}^n x_i = 360$

➤ **Numerical Example:**

✧ **Calculate the arithmetic mean for the following data given below:**

◆ Using formula of **direct method** of arithmetic mean for grouped data:

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}, \quad n = \sum_{i=1}^n f_i$$

The weight recorded to the nearest grams of 60 apples picked out at random from a consignment are given below:

106	107	76	82	109	107	115	93	187	95	123	125
111	92	86	70	126	68	130	129	139	119	115	128
100	186	84	99	113	204	111	141	136	123	90	115
98	110	78	185	162	178	140	152	173	146	158	194
148	90	107	181	131	75	184	104	110	80	118	82

Weight (grams)	Frequency
65----84	09
85----104	10
105----124	17
125----144	10
145----164	05
165----184	04
185----204	05

Solution:

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	$f_i x_i$
65----84	$(65 + 84)/2 = 74.5$	09	$9 \times 74.5 = 670.5$
85----104	94.5	10	945.0
105----124	114.5	17	1946.5
125----144	134.5	10	1345.0
145----164	154.5	05	772.5
165----184	174.5	04	698.0
185----204	194.5	05	972.5
		$\sum_{i=1}^n f_i = 60$	$\sum_{i=1}^n f_i x_i = 7350.0$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{7350.0}{60} = 122.5 \text{ grams } \textbf{(Answer).}$$

◆ Using formula of **short cut method** of **arithmetic mean** for **grouped data**:

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i D_i}{\sum_{i=1}^n f_i}, \quad n = \sum_{i=1}^n f_i$$

Where $D_i = X_i - A$ and A is the provisional or assumed mean

- **Geometric Mean:** “The n^{th} root of the product of “ n ” positive values is called geometric mean”

$$\text{Geometric Mean} = \sqrt[n]{\text{product of "n" positive values}}$$

The following are the formulae of geometric mean:

Ungrouped data	Grouped data
$G = \text{Antilog}\left(\frac{\sum \log x}{n}\right)$	$G = \text{Antilog}\left(\frac{\sum f \log x}{n}\right)$; Here $n = \sum f$

- ❖ **Numerical example of geometric Mean for both grouped and ungrouped data:**
- **Calculate the geometric mean for the following the marks obtained by 9 students are given below:**

x_i	45	32	37	46	39	36	41	48	36
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- ✧ Using formula of geometric mean for ungrouped data:

$$n = 9$$

x_i	$\log x_i$
45	$\log 45 = 1.65321$
32	1.50515
37	1.56820
46	1.66276
39	1.59106
36	1.55630
41	1.61278
48	1.62124
36	1.55630
	$\sum_{i=1}^n \log x_i = 14.38700$

$$G.M = \text{anti} - \log \left(\frac{\sum_{i=1}^n \log x_i}{n} \right)$$

$$G.M = \text{anti} - \log \left(\frac{14.38700}{9} \right)$$

$$G.M = \text{anti} - \log (1.59856)$$

$$G.M = 39.68 \quad (\text{Answer}).$$

- Given the following frequency distribution of weights of 60 apples, calculate the geometric mean for grouped data.

Weights (grams)	65--84	85--104	105--124	125--144	145--164	165--184	185--204
Frequency	09	10	17	10	05	04	05

$$G.M = anti - \log \left(\frac{\sum_{i=1}^n f_i \log x_i}{\sum_{i=1}^n f_i} \right)$$

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	$\log x_i$	$f_i \log x_i$
65----84	$(65 + 84)/2 = 74.5$	09	1.8722	16.8498
85----104		10	1.9754	19.7540
105----124		17	2.0589	35.0013
125----144	94.5	10	2.1287	21.2870
145----164	114.5	05	2.1889	10.9445
165----184	134.5	04	2.2418	8.9672
185----204	154.5	05	2.2889	11.4445
	174.5			
	194.5			
		$\sum_{i=1}^n f_i = 60$		$\sum_{i=1}^n f_i \log x_i =$ 124.2483

$$G.M = anti - \log \left(\frac{\sum_{i=1}^n f_i \log x_i}{\sum_{i=1}^n f_i} \right)$$

$$= anti - \log \left(\frac{124.2483}{60} \right)$$

$$= anti - \log (2.0708)$$

$$G.M = 117.7$$

grams

(An
swer).

◆ **Harmonic Mean:** “The reciprocal of the Arithmetic mean of the reciprocal of the values is called Harmonic mean”

$$\text{Harmonic Mean} = \text{reciprocal of} \left(\frac{\text{Sum of reciprocal of the values}}{\text{The number of values}} \right)$$

The following are formulae of harmonic mean:

Ungrouped data	Grouped data
$H = \frac{n}{\sum \left(\frac{1}{x} \right)}$	$H = \frac{n}{\sum \left(\frac{f}{x} \right)}$; Here $n = \sum f$

❖ **Numerical example of harmonic Mean for both grouped and ungrouped data:**

➤ **Calculate the harmonic mean for the following the marks obtained by 9 students are given below:**

x_i	45	32	37	46	39	36	41	48	36
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✧ Using formula of harmonic mean for ungrouped data:

$$n = 9$$

x_i	$1/x_i$
45	0.02222
32	0.03125
37	0.02702
46	0.02173
39	0.02564
36	0.02777
41	0.02439
48	0.02083
36	0.02777
$\sum_{i=1}^n \frac{1}{x_i} = 0.22862$	

$$H.M = \frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i} \right)}$$

$$H.M = \frac{9}{0.22862}$$

$$H.M = 39.36663 \quad (\text{Answer}).$$

➤ Given the following frequency distribution of weights of 60 apples, calculate the harmonic mean for grouped data.

Weights (grams)	65--84	85--104	105--124	125--144	145--164	165--184	185--204
Frequency	09	10	17	10	05	04	05

$$H.M = \frac{\sum_{i=1}^n f_i}{\sum_{i=1}^n \left(\frac{f_i}{x_i} \right)}$$

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	f_i/x_i
65----84	$(65+84)/2 = 74.5$	09	0.12081
85----104	94.5	10	0.10582
105----124	114.5	17	0.14847
125----144	134.5	10	0.07435
145----164	154.5	05	0.03236
165----184	174.5	04	0.02292
185----204	194.5	05	0.02571
		$\sum_{i=1}^n f_i = 60$	$\sum_{i=1}^n \frac{f_i}{x_i} = 0.53044$

$$H.M = \frac{\sum_{i=1}^n f_i}{\sum_{i=1}^n \left(\frac{f_i}{x_i} \right)} = \frac{60}{0.53044} = 113.11 \text{ grams} \quad (\text{Answer}).$$

- ❖ **Median:** “when the observation are arranged in ascending or descending order, then a value, that divides a distribution into equal parts, is called median”

Median in case of Ungrouped Data	
In this case we first arrange the observations in increasing or decreasing order then we use the following formulae for Median:	
If “n” is odd	Median = size of $\left(\frac{n+1}{2} \right)$ th observation
If “n” is even	Median = $\frac{\text{size of } \left\{ \left(\frac{n}{2} \right)\text{th} + \left(\frac{n}{2} + 1 \right)\text{th} \right\} \text{ observation}}{2}$

- **Numerical example of median for both grouped and ungrouped data:**

If “n” is odd

$$\text{Median} = \text{size of } \left(\frac{n+1}{2} \right)\text{th observation}$$

- **Calculate the median for the following the marks obtained by 9 students are given below:**

Arrange the

x_i	45	32	37	46	39	36	41	48	36
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 data in